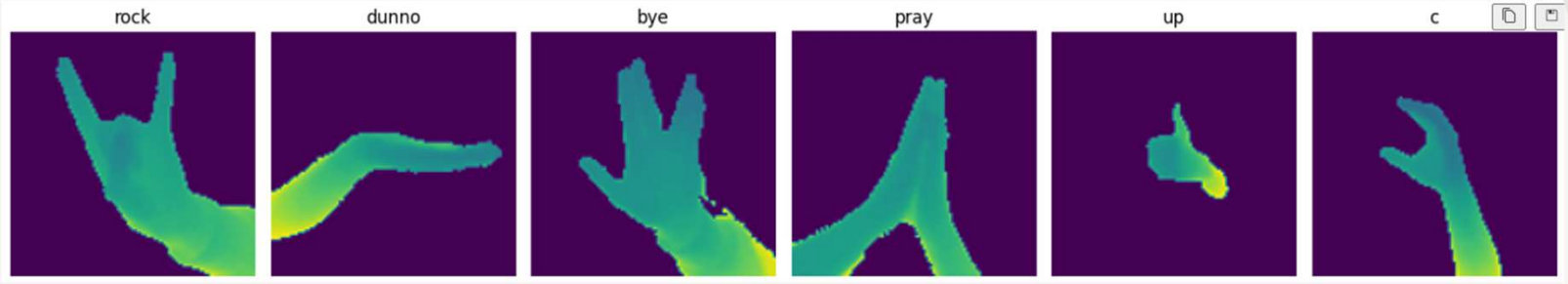
Projekt 2

BC. Juraj Kováč

BC. Viktor Hrúz





Rozpoznávanie gest ruky z hĺbkovej mapy

Serial CNN

```python
def forward(self, x):
    x = self.layer1(x)
    x = self.layer2(x)
    x = self.layer3(x)
    x = x.view(x.size(0), -1)
    x = self.fc(x)
    return x
```

```python
def __init__(self, in_channels, num_classes):
    super(SerialNet, self).__init__()

    self.layer1 = nn.Sequential(
        nn.Conv2d(in_channels, 32, kernel_size=3, padding=1),
        nn.BatchNorm2d(32),
        nn.PReLU(),
        nn.MaxPool2d(kernel_size=2, stride=2)  # 224x224 -> 112x112
    )

    self.layer2 = nn.Sequential(
        nn.Conv2d(32, 64, kernel_size=5, padding=2),
        nn.BatchNorm2d(64),
        nn.PReLU(),
        nn.MaxPool2d(kernel_size=2, stride=2)  # 112x112 -> 56x56
    )

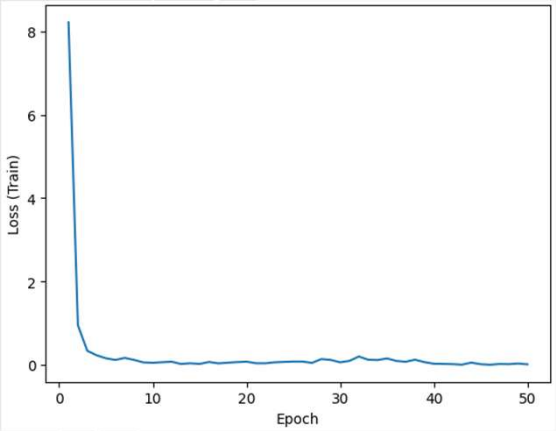
    self.layer3 = nn.Sequential(
        nn.Conv2d(64, 128, kernel_size=3, padding=1),
        nn.BatchNorm2d(128),
        nn.PReLU(),
        nn.MaxPool2d(kernel_size=2, stride=2)  # 56x56 -> 28x28
    )

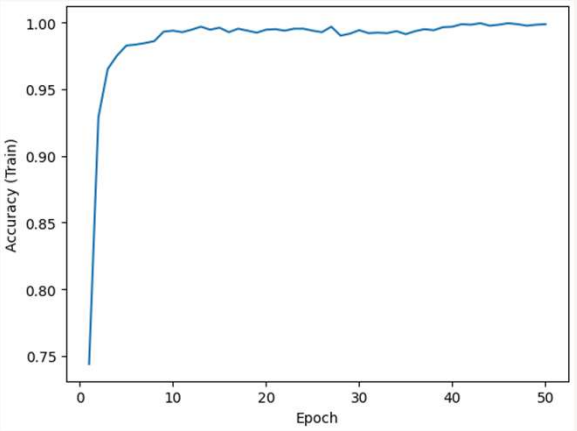
    self.fc = nn.Sequential(
        nn.Linear(128 * 28 * 28, 256),
        nn.PReLU(),
        nn.Dropout(0.5),
        nn.Linear(256, num_classes)
    )
```

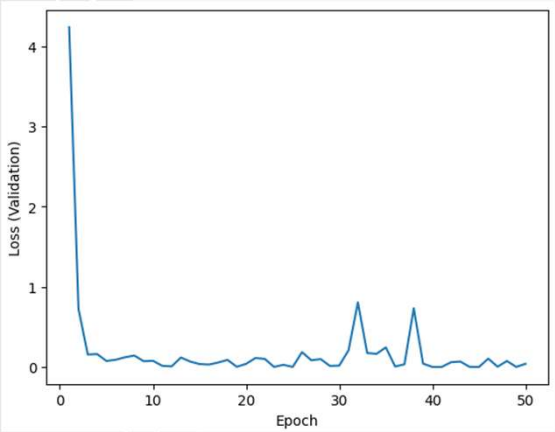
Serial CNN

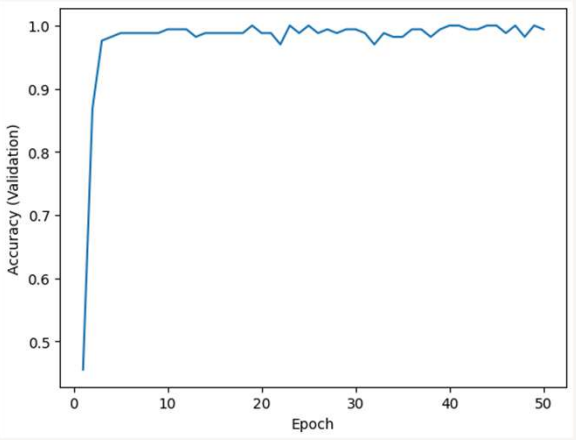
- 560 obrázkov v skupine
  - Train :448
  - Test:84
  - Val:28
- Learning rate:0,001
- Batch size: 100
- Grayscale

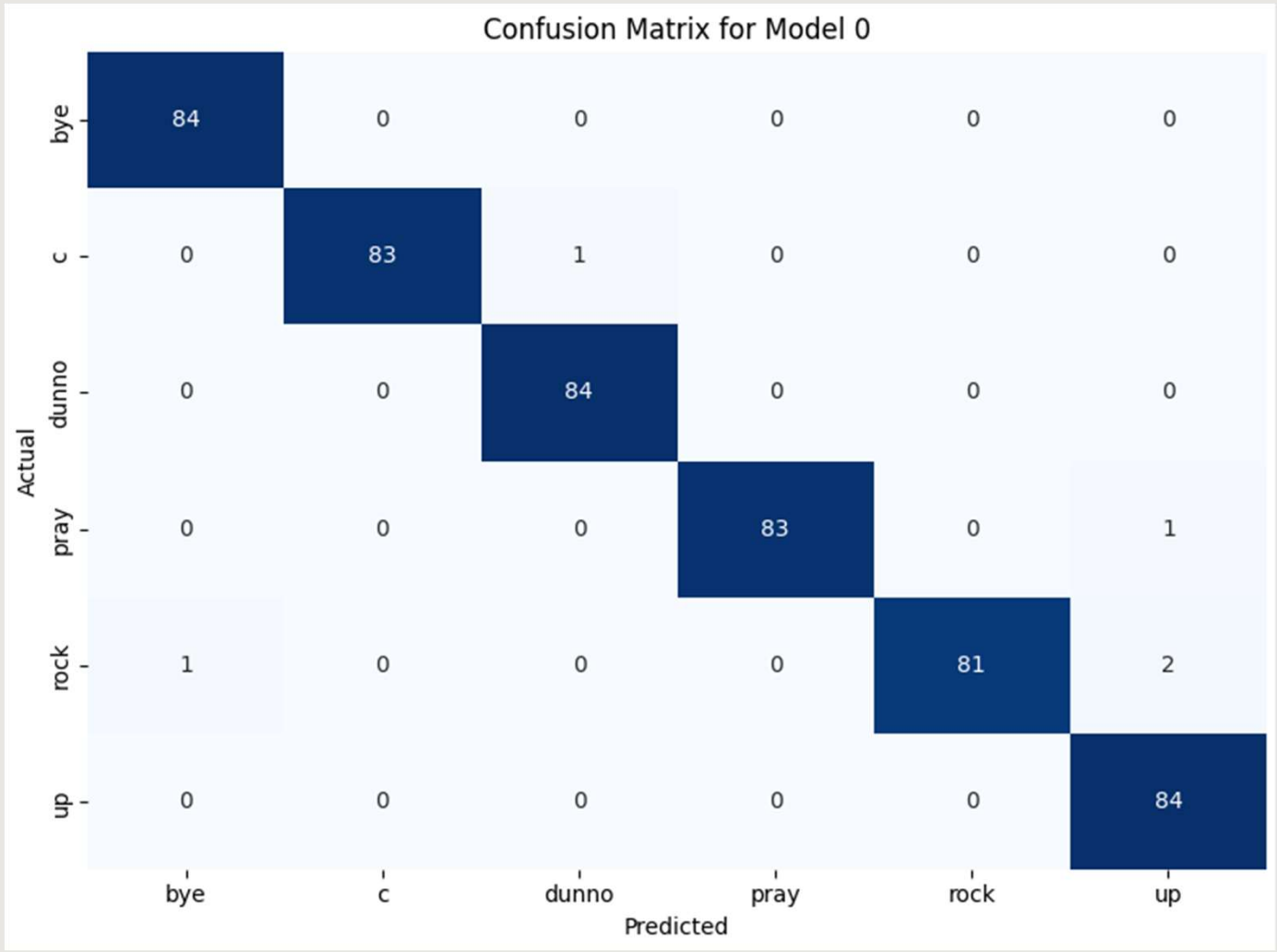
Testovacia presnosť: 99,01%










Confusion Matrix for Model 0
bye
c
dunno
pray
rock
up
Actual
84 0 0 0 0 0
0 83 1 0 0 0
0 0 84 0 0 0
0 0 0 83 0 1
1 0 0 0 81 2
0 0 0 0 0 84
bye
c
dunno
pray
rock
up
Predicted



Rozpoznávanie gest ruky z RGB obrazu

Serial CNN

```python
def forward(self, x):
    x = self.layer1(x)
    x = self.layer2(x)
    x = self.layer3(x)
    x = x.view(x.size(0), -1)
    x = self.fc(x)
    return x
```

```python
def __init__(self, in_channels, num_classes):
    super(SerialNet, self).__init__()

    self.layer1 = nn.Sequential(
        nn.Conv2d(in_channels, 32, kernel_size=3, padding=1),
        nn.BatchNorm2d(32),
        nn.PReLU(),
        nn.MaxPool2d(kernel_size=2, stride=2)  # 224x224 -> 112x112
    )

    self.layer2 = nn.Sequential(
        nn.Conv2d(32, 64, kernel_size=5, padding=2),
        nn.BatchNorm2d(64),
        nn.PReLU(),
        nn.MaxPool2d(kernel_size=2, stride=2)  # 112x112 -> 56x56
    )

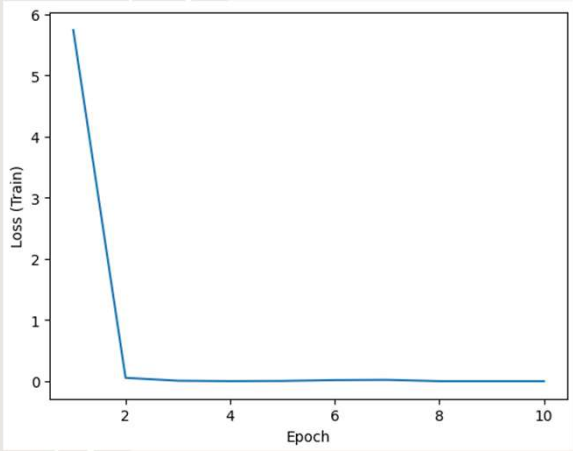
    self.layer3 = nn.Sequential(
        nn.Conv2d(64, 128, kernel_size=3, padding=1),
        nn.BatchNorm2d(128),
        nn.PReLU(),
        nn.MaxPool2d(kernel_size=2, stride=2)  # 56x56 -> 28x28
    )

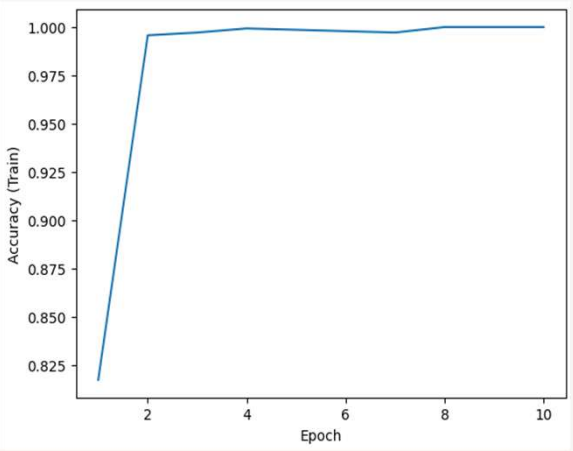
    self.fc = nn.Sequential(
        nn.Linear(128 * 28 * 28, 256),
        nn.PReLU(),
        nn.Dropout(0.5),
        nn.Linear(256, num_classes)
    )
```

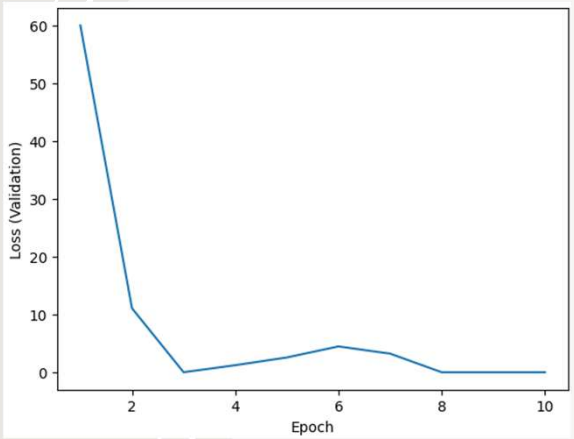
Serial CNN

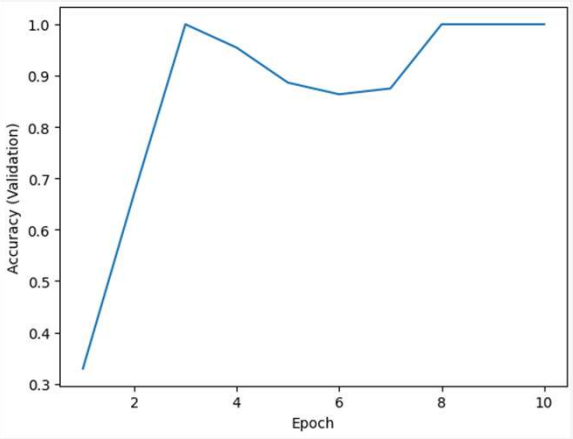
- 440 obrázkov v skupine
  - Train : 352
  - Test:66
  - Val:22
- Learning rate:0,001
- Batch size: 50

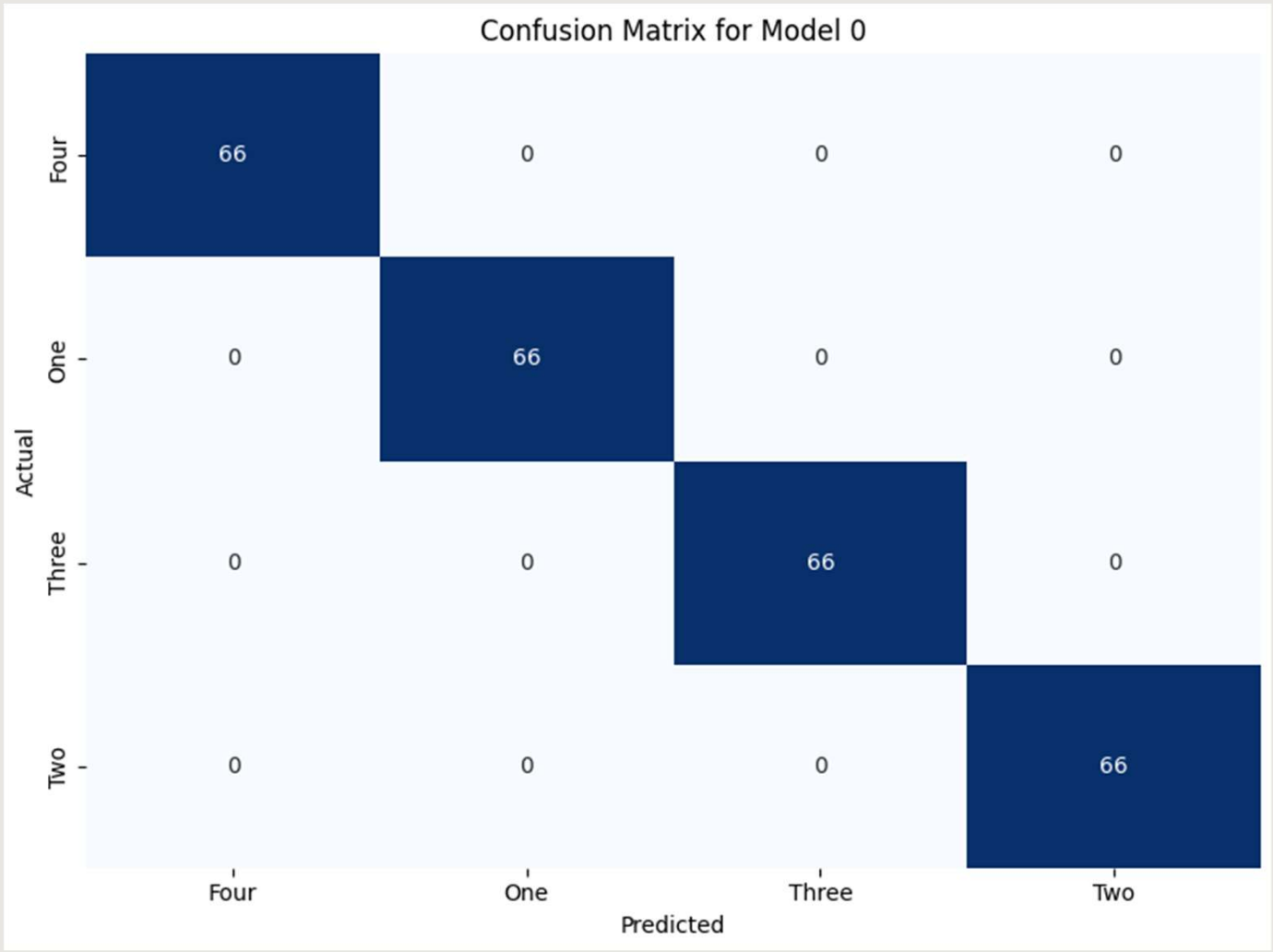
Testovacia presnosť: 100%










Confusion Matrix for Model 0
Four
One
Three
Two
Actual
66
0
0
0
0
66
0
0
0
0
66
0
0
0
0
66
Four
One
Three
Two
Predicted

BONUS

Rozšírenie datasetu

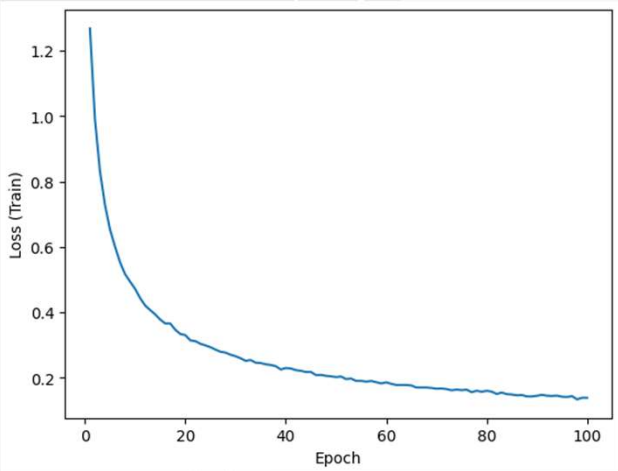
Zo 440 obrázkov na skupinu na 1800

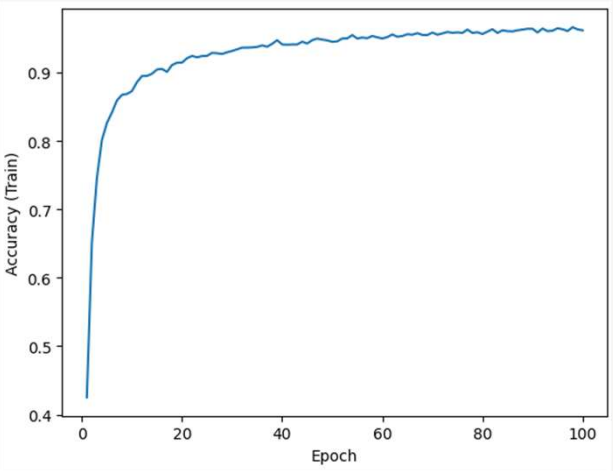


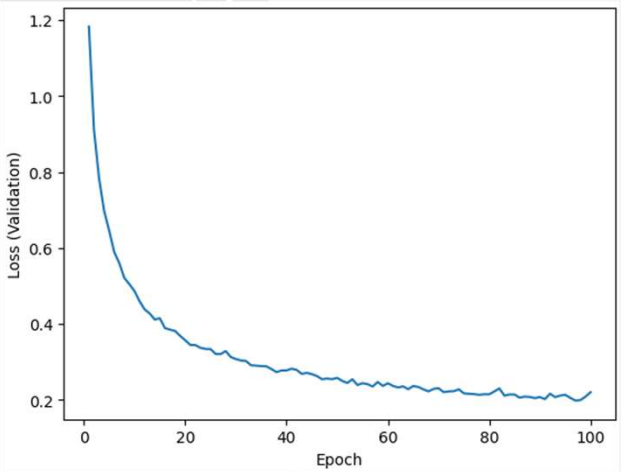
ResNet18

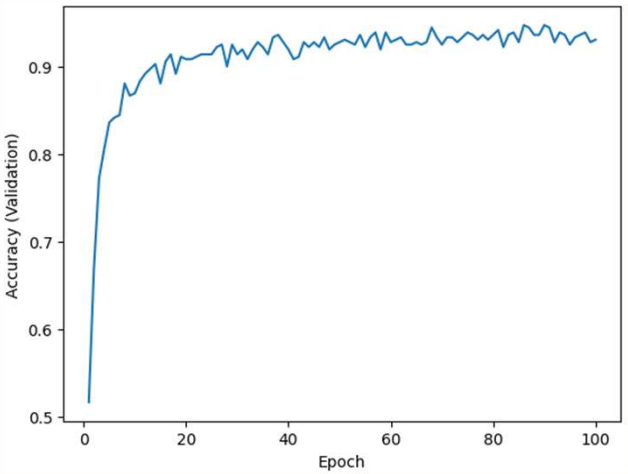
- 1800 obrázkov v skupine
  - Train: 1440
  - Test:270
  - Val:90
- Learning rate:0,001
- Batch size: 200

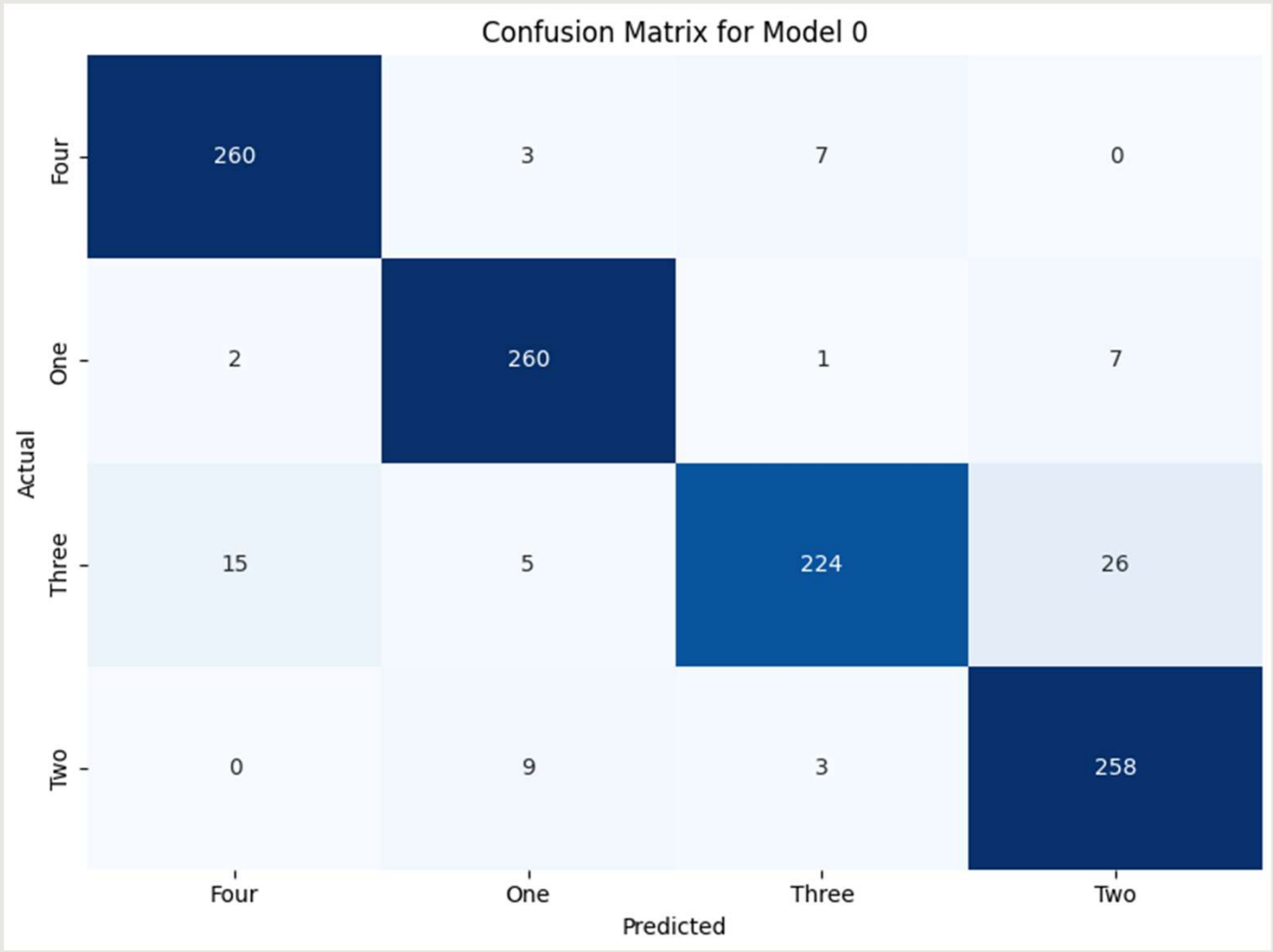
Testovacia presnosť: 92,78%










Confusion Matrix for Model 0
Four
One
Three
Two
Actual
260
3
7
0
2
260
1
7
15
5
224
26
0
9
3
258
Four
One
Three
Two
Predicted

Test na kamere

Video v prílohe

Ďakujeme za pozornosť